Question 1

- Name three lossy compression methods

- *The three lossy compressions are JPEG, MPEG, and MP3*.

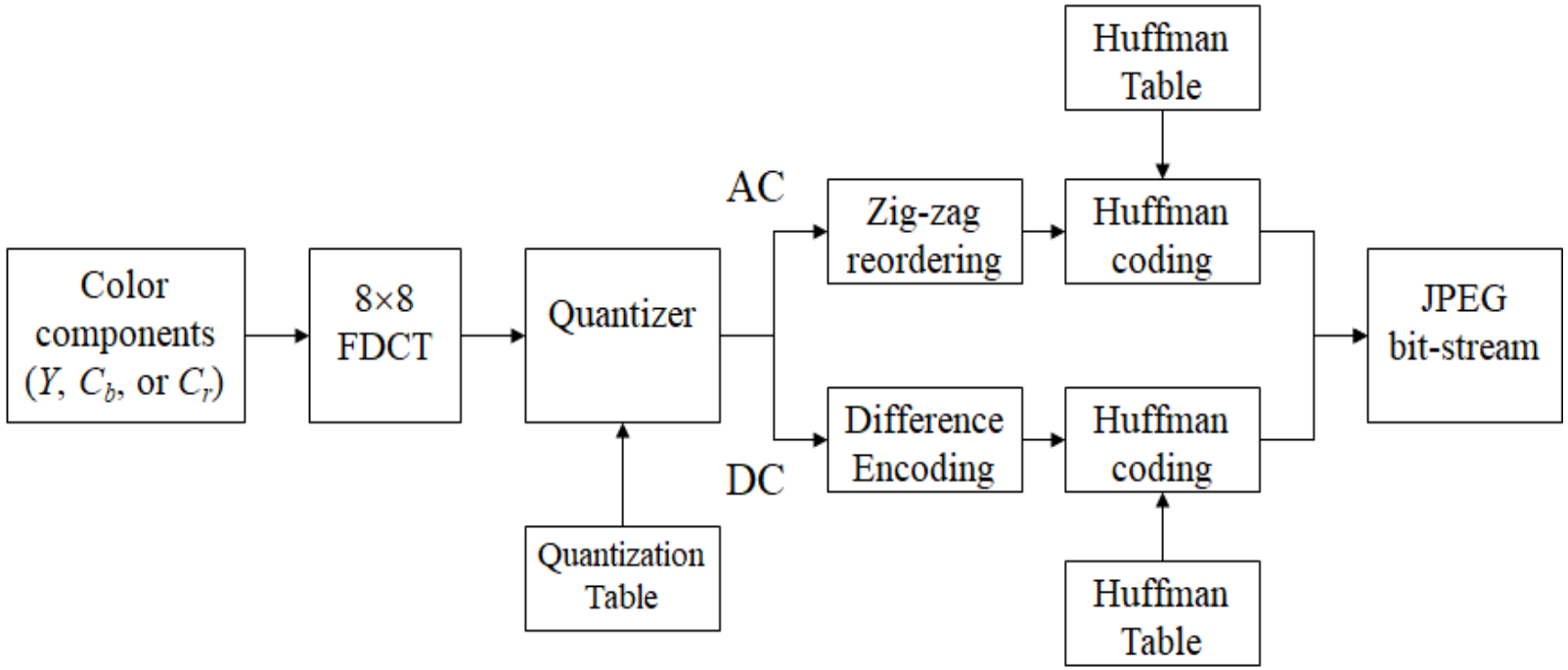
Question 2

- When would you use JPEG? When would you use MPEG?

- *JPEG is used to compress images while MPEG is used to compress video.*

Question 3

- Describe various stages of the JPEG algorithm



RGB to YC_bC_r conversion

- Instead of dealing with the pixels in the RGB colour space, JPEG converts them to the YC_bC_r colour space.
- The YC_bC_r colour space has three components per pixel:
  - the luminance, or the brightness of the pixel, Y,
  - the amount of blue light in the pixel, C_b, and,
  - the amount of red light in the pixel, C_r.
- The Y component is called the luma component and the C_b & C_r components are known as the chroma components.

Chroma Subsampling

- The human eye is better at dealing with the brightness or luminescence than colours.
- So, the separation of the luma (Y) and chroma (Cb & Cr) components enables us to throw away some of the information relating to the colour, without significant loss of the overall visual quality of the image.

Question 4

- Why is the discrete cosine transform needed in JPEG?

- *DCT changes the 64-pixel values in each block so that the relative relationships between pixels are kept but the redundancies are removed.* 保留像素值之间的相对关系，去除冗余

Discrete Cosine Transform (DCT)

- Given a signal that is a function of time, it can be decomposed into sinusoidal components of varying frequencies that make it up. You have heard about Fourier transform, which is one of the frequency transforms that can do this. DCT is a Fourier related frequency transform, which is what JPEG uses.

$$\text{FDCT}: S_{vu} = \frac{1}{4} C_u C_v \sum_{x=0}^{7} \sum_{y=0}^{7} s_{yx} \cos \frac{(2x+1)u\pi}{16} \cos \frac{(2y+1)v\pi}{16}$$

Question 5

- Why we change RGB space to YCbCr

- The retina of an eye has two major types of photoreceptors, *rods* and *cones*. The rods are sensitive to the brightness of light and the cones to the colour.
- The number of rods in the retina massively outnumber the number of cones. As a result, a human eye is able to tell apart different levels of light intensity better than they can discriminate between different colours.

Question 6

- In JPEG, what is the function of blocking?

- *Blocking is the act of dividing the image into 8×8 blocks in order to reduce the number of calculations.*

Question 7

- How does quantisation contribute to compression?

- *Quantization of the table reduces the number of bits needed for encoding each value.*

Last year's class test questions

- **Question 1:**

low-level = Input is picture, output is picture too, some basic operation
high-level = Input is picture, output is meaningful understanding of the image

- Explain the difference between the high- and low-level image processing tasks. [5 Marks]

- **Question 2:**

- What is the difference between lossless compression and lossy compression? Explain with examples. [5 Marks]

一个保留了数据的完整性
一个有数据的缺失。

- **Question 3:**

- A Digital Source X has five symbols x1 , x2, x3, x4, and x5 with P(x1) = 0.4, P(x2) = 0.19, P(x3) = 0.16, P(x4) = 0.15, and P(x5) = 0.1. Construct a Huffman code for X and calculate the efficiency of the code.

- [10 Marks]

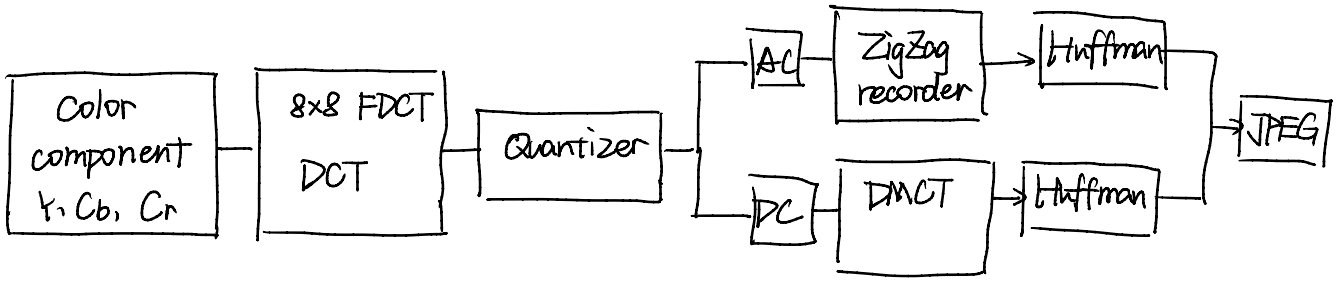
$$L = \sum_{i=1}^{5} P(x_i) n_i$$

$$L = 0.4(1) + (0.19 + 0.16 + 0.15 + 0.1)(3) = 2.2$$

$$\eta = \frac{H(X)}{L}$$

Cont'd

- **Question 4:**



- Give the block diagram of the JPEG algorithm. [5 Marks]

- **Question 5:**

- What does the general sampling (Nyquist) theorem state? [5 Marks]

Nyquist 抽样定理：

像素尺寸 < 原始图像中最细尺寸的一半

灰度级亮度 < 原始图像最小色调变化的一半